

Python Coding Test

Instructions

- Write your solutions in Python.
 - Each question corresponds to a specific concept or category listed below.
 - Ensure your code is correct, well-structured, and follows Python best practices.
 - Submit your solutions in a single `.py` file.
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1. Python Intro

Question: Write a Python program that prints "Hello, Python!" to the console.

2. Python Get Started

Question: Create a Python script that accepts a user's name as input and prints a personalized welcome message, e.g.,
"Welcome to Python, [Name]!".

3. Python Syntax

Question: Write a Python program to calculate the area of a rectangle. Use meaningful variable names for length and width, and follow Python's indentation rules.

4. Python Comments

Question: Write a Python script that calculates the sum of the first 10 positive integers. Add comments to explain each step of the code.

5. Python Variables

Question: Create a Python program that swaps the values of two variables without using a temporary variable.

6. Python Data Types

Question: Write a Python script that declares variables of the following data types: integer, float, string, and boolean. Print the type of each variable using the `type()` function.

7. Python Numbers

Question: Write a Python program that generates a random integer between 1 and 100 (inclusive). Use the `random` module.

8. Python Casting

Question: Write a Python script that accepts a floating-point number as input, converts it to an integer, and prints both the original float and the converted integer.

9. Python Strings

Question: Write a Python program that accepts a sentence as input and prints:

1. The sentence in uppercase.
 2. The sentence in lowercase.
 3. The number of characters in the sentence.
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10. Python Booleans

Question: Write a Python script that checks if a number is even or odd and prints `True` if it's even, otherwise `False`.

11. Python Operators

Question: Write a Python program that calculates the result of the following expression: $(50 + 25) * 2 - 10 / 5$. Print the result.

12. Python Lists

Question: Write a Python program that creates a list of the first 5 odd numbers. Add one more odd number to the list and print the updated list.

13. Python Tuples

Question: Write a Python program that creates a tuple with the numbers 1 to 5. Attempt to change the second element of the tuple and observe what happens. Add a comment explaining the result.

14. Python Sets

Question: Write a Python script that creates two sets:

- Set A: {1, 2, 3, 4}
- Set B: {3, 4, 5, 6} Find and print the following:

1. The union of the sets.
 2. The intersection of the sets.
 3. The difference between Set A and Set B.
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Submission Guidelines

- Ensure all your code runs without errors.
- Add comments where necessary to explain your logic.
- Include all solutions in a single `.py` file, with each solution clearly separated by comments or headers.

Good luck!